

Robust DNA Data Storage via Block-by-Block Synthesis: The PEARL-DNA Architecture

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As global data generation accelerates exponentially, the rapidly growing field of DNA data storage provides unprecedented advantages in scalability, stability, and efficiency. While various research groups have demonstrated the potential of *in vitro* DNA data storage, the integration of biological components introduces complex sources of errors throughout the data processing chain, potentially compromising data integrity and reliability. The PEARL-DNA project establishes a proof-of-concept system for fast, cost-efficient, and reliable block-by-block high-throughput DNA assembly. It implements an end-to-end pipeline encompassing synthesis, amplification, storage, sequencing, and decoding.

The PEARL-DNA technology is founded on the principle of block-by-block synthesized DNA. This approach offers several benefits, notably in adherence to ethics guidelines by ensuring all oligos are constructed from a priori-known, verified biosafe DNA segments. Furthermore, knowing the precise sequence of DNA segments that form an oligonucleotide is a key advantage that can be exploited during data retrieval.

Specifically, the PEARL-DNA system includes DNA synthesis reactions optimized to operate under conditions suitable for small reaction volumes in automated liquid handling systems. By combining standard droplet generation technologies, the system achieves efficient logical reagent mixing. Furthermore, the system includes the best room-temperature storage solution, by desiccating DNA and maintaining it under a dry and inert atmosphere. This technology has been adapted to offer higher density and to support very high-throughput automation. Finally, the system integrates a multi-layered error correction strategy to ensure reliable data retrieval. This includes a combination of LDPC codes, RS codes, and interleaving in the digital domain. Crucially, during decoding, we apply a tailored alignment algorithm that exploits the unique block-by-block structure, effectively converting challenging insertion and deletion errors into substitution errors. These are then corrected by the LDPC/RS decoding steps.

We validated the PEARL-DNA architecture on two distinct datasets. Both datasets were synthesized, underwent simulated storage over approximately 500 years at 25°C, and were sequenced. In both cases, using only the tailored alignment algorithm, the data was recovered with a final bit error rate of approximately 0.5%. Our *in silico* validation of the entire pipeline demonstrated a perfect reconstruction with up to a 20% substitution error rate of the pre-defined DNA blocks at a code rate of 0.6. In summary, PEARL-DNA shows that block-by-block DNA data storage functions effectively at scale and exhibits robust resilience.